

Creating a Program to Support Registering and Reporting Clinical Trials at Johns Hopkins University

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Abstract

Problem

The Food and Drug Administration Amendments Act of 2007 (FDAAA) and the National Institutes of Health (NIH) require that many clinical trials register and report results on ClinicalTrials.gov. Noncompliance with these policies denies research participants and scientists access to potentially relevant findings and could lead to monetary penalties or loss of funding. After discovering hundreds of potentially noncompliant trials affiliated with the institution, the Johns Hopkins University School of Medicine (JHUSOM) sought to develop a program to support research teams with registration and reporting requirements.

Approach

JHUSOM conducted a baseline assessment of institutional compliance in 2015, launched the ClinicalTrials.gov Program in June 2016, and expanded the program to the Sidney Kimmel Comprehensive Cancer Center in April 2018. The program is innovative in its comprehensive approach, and it was among the first to bring a large number of trials into compliance.

Outcomes

From September 2015 to September 2020, JHUSOM brought completed and ongoing trials into compliance with FDAAA and NIH policies and maintained

almost perfect compliance for new trials. During this period, the proportion of trials potentially noncompliant with the FDAAA decreased from 44% (339/774) to 2% (32/1,304).

Next Steps

JHUSOM continues to develop and evaluate tools and procedures that facilitate trial registration and results reporting. In collaboration with other academic medical centers, JHUSOM plans to share resources and to identify and disseminate best practices. This report identifies practical lessons for institutions that might develop similar programs.

Problem

ClinicalTrials.gov is an online registry and results database of clinical trial information that is operated by the National Library of Medicine (NLM). Since 2005, journals adopting the recommendations of the International Committee of Medical Journal Editors will consider trial manuscripts for publication only if registration of the trial occurred before the first patient was enrolled.¹ The Food and Drug Administration Amendments Act of 2007 (FDAAA) requires that all applicable clinical trials be registered within 21 days of enrollment and that results be reported within 1 year of trial completion.² The FDAAA authorizes the Food and Drug Administration (FDA) to issue civil monetary penalties for noncompliance,

which are currently \$12,316 per trial per day. The Final Rule, which took effect January 18, 2017, clarified and expanded the FDAAA, with organizations expected to be in compliance by April 18, 2017.^{3,4} At the same time, the National Institutes of Health (NIH) issued broader requirements that apply to all trials funded by the NIH.⁵ Research groups that do not comply with the NIH policy can lose their funding.⁴

In March 2014, several events prompted the Johns Hopkins University School of Medicine (JHUSOM) to initiate an assessment of institutional compliance with the requirements for clinical trial registration and results reporting. A principal investigator (PI) affiliated with the school had received a letter from the FDA citing noncompliance with the FDAAA, and other investigators were seeking assistance using ClinicalTrials.gov. Thus, JHUSOM vice dean of clinical investigation (D.E.F.) asked a project manager (PM; A.K.), to assess compliance at the institution.

At the time, little published literature described how to build a compliance program at an academic medical center.

The PM attended an NLM workshop designed to help institutions understand and use ClinicalTrials.gov.⁶ In June 2015, the PM obtained administrative control of all trials registered by JHUSOM in the Protocol Registration and Results System (PRS), the online system that allows the institution to register, update, and report results on ClinicalTrials.gov, and ran a baseline report of compliance.

In the PRS, registered trials are assigned a National Clinical Trial (NCT) number and are referred to as “records.” The baseline report in September 2015 identified 339 of 774 trials (44%) as being potentially noncompliant with the FDAAA (i.e., “problem records”); of these, 107 trials were identified as “Late results—per FDAAA” (late results), which are trials that have not been reported within 1 year of the date of last data collection for all primary outcome measures. At the time, we estimated that civil monetary penalties of approximately \$10,000 per trial per day could result in a potential institutional liability of over \$1,000,000 per day.

To ensure that the institution remained a responsible member of the research

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community, to alleviate potential financial liability, and because of the specialized knowledge required to understand the relevant clinical trial registration and results reporting requirements, we established a formal program to support investigators in compliance with these requirements. Here, we share the steps we took and the lessons we learned while establishing the ClinicalTrials.gov Program at JHUSOM for institutions that might develop similar programs.

Approach

The timeline shown in Figure 1 lists steps we took while establishing the ClinicalTrials.gov Program and the staffing levels entailed.

Evaluating processes at peer institutions

Before developing our program, the PM joined the Clinical Trials Registration and Results Reporting Taskforce.⁷ Through this taskforce, the PM communicated with peer institutions to discuss their policies, procedures, and resources related to trial registration and results reporting. The taskforce continues to meet monthly and allows institutions to share information through its website.

Accessing trial records

Colleagues at peer institutions described the importance of internal collaborations with the institutional review board (IRB) and other relevant stakeholders. Before launching the program, the JHUSOM IRB director granted the PM read-only access to all studies, which allowed the PM to verify current study status, run reports, and contact investigators as needed.

Launching the ClinicalTrials.gov Program at JHUSOM

Highlighting potential financial liability, the PM made a business case

to leadership to secure institutional and financial support through the Clinical and Translational Science Award (CTSA). JHUSOM assigned the PM to work on trial registration and results reporting. JHUSOM hired a clinical research compliance specialist and launched the program in June 2016 with the following objectives:

- Develop institutional ClinicalTrials.gov policies
- Develop a process for communicating with investigators
- Identify and reduce problem records
- Develop and implement proactive compliance measures
- Train investigators to meet trial registration and results reporting requirements
- Expand the program across JHUSOM

Institutional policies and procedures

The program revised an existing JHUSOM Organization Policy on Registration of Clinical Trials to meet the requirements in the Final Rule.⁴ After approval by the vice dean, we posted the updated policy to the institutional website.

In addition, we revised the IRB application to collect the details necessary to ensure that all clinical trials meeting the statutory requirements were appropriately identified. Program staff created standard operating procedures to ensure institutional uniformity of activities such as creating user accounts within the PRS, transferring records between institutions, and communicating requirements to PIs and departmental leadership. Today, our program regularly reviews JHUSOM policies to ensure they reflect current regulations and guidelines.

Communication processes

In 2016, the program staff began to communicate the requirements for registration and results reporting to investigators who might be noncompliant with the FDAAA. The vice dean also informed department directors of their financial responsibility for any monetary penalties levied against the institution.

For records with late results or other problems, program staff first notified investigators from the program's email account. Program staff explained the problem and described steps to take to remedy it. If the investigator did not address the problem, the staff sent a second email and copied the department director. Occasionally, the program staff sent a third email copying the vice dean. Program staff also made phone calls and personal visits to meet investigators when necessary.

Eliminating late results

Our first steps in 2016 involved reporting late results. We found that some trials that were identified in the PRS as late results were still actively recruiting. For these trials, the program staff worked with the investigators to update the records to indicate that results were expected in the future.

We also discovered that several records identified as late results had not enrolled any participants; updating the trial status to "withdrawn" eliminated the requirement to report results.

We learned from peer institutions to start with protocols that had been late the longest. To collect the required information, we first reached out to PIs and then to anyone identified in the ClinicalTrials.gov record, IRB study team members, and manuscript coauthors. Some PIs and staff members

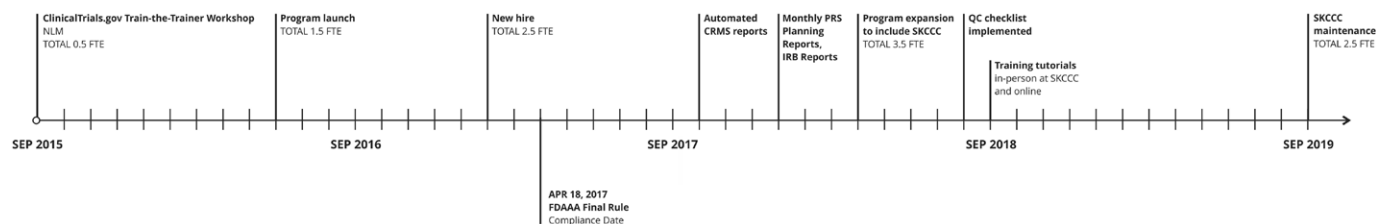


Figure 1 ClinicalTrials.gov program development timeline, Johns Hopkins University School of Medicine, begun 2015, ongoing. Key events are shown, including changes in staffing during stages of program development and in relation to the compliance date for the FDAAA 2007 Final Rule. Abbreviations: NLN, National Library of Medicine; FTE, full-time equivalent; FDAAA, Food and Drug Administration Amendments Act; CRMS, Clinical Research Management System; PRS, Protocol Registration and Results System; IRB, institutional review board; SKCCC, Sidney Kimmel Comprehensive Cancer Center; QC, quality control.

had transferred departments, left the institution, or retired. We retrieved historical data for 10 trials from off-site storage, and we used the IRB termination report (i.e., a report submitted by the PI) to enter results into ClinicalTrials.gov for those trials.

Lastly, we communicated to investigators that the April 2017 Final Rule compliance date was the deadline for submitting all required information. We received a robust response before the deadline; 35% (31/88) of the trial results we submitted in 2017 were within ± 2 weeks of the deadline.

Preventing problems prospectively

After hiring a second staff member in 2017, we began to use the PRS planning report to prospectively identify records requiring results reporting. We began to reach out to investigators 4 months before the results were due. If the anticipated completion date had changed, we assisted investigators in updating the record and reminded them to maintain the record's accuracy.

Many investigators require support throughout the registration and results reporting process. We provide this support at no cost to the investigators. We also perform tasks on a fee-for-service basis. Since 2017, 6 investigators have asked us to enter results for 9 trials, which required 64 hours of effort, for which we charged \$50 per hour.

In addition to late results, the PRS identifies problems such as “not recently updated,” “missing FDAAA information,” “incomplete results—per FDAAA,” and “PRS review comments.” A single record could have multiple problems. Today, the program reviews problem records daily, enabling us to quickly recognize new problems. In addition, we developed a comprehensive checklist that we use to help orient new program staff and to ensure consistency among administrators within our institution. The ClinicalTrials.gov PRS team at the NIH is helpful with challenging cases.

Although JHUSOM does not mandate training for investigators, we developed voluntary in-person training sessions to help investigators maintain compliance and avoid penalties. These sessions are available to departments upon request, and recordings are available through YouTube and on our website.

Monitoring compliance

At JHUSOM, participant enrollment in clinical trials is recorded in the Clinical Research Management System (CRMS), an online reporting platform. In collaboration with the institution's IRB, we automated the reports created to ensure that the information in ClinicalTrials.gov is accurate and up to date. These reports include one that identifies trials that began enrollment without citing the NCT number in either the IRB application or the CRMS. We ran a baseline report in August 2017 and identified 149 trials that we investigated until the NCT number was entered. Today, we reach out to investigators of new trials to ensure prompt registration.

Expanding the program

In addition to the JHUSOM PRS, other Johns Hopkins University entities have PRS accounts. In April 2018, the Sidney Kimmel Comprehensive Cancer Center at Johns Hopkins (SKCCC) hired a clinical research compliance specialist to focus exclusively on SKCCC records. SKCCC already had procedures for clinical trial research operations, and we collaborated with their Clinical Research Office to facilitate communication and encourage cooperation. The staff within this office have established relationships with investigators and historical knowledge about trials and staff turnover. Just as JHUSOM began by addressing existing problem records, in its first year, the SKCCC program aimed toward retrospective compliance by addressing problem records and assigning appropriate individuals as record owners or access list members.

Outcomes

Compliance with the FDAAA and NIH policies increased for completed and ongoing trials from September 2015 (baseline) to September 2020 (4 years after we introduced the ClinicalTrials.gov Program at JHUSOM), as shown in Figure 2.

In the annual report run in September 2016 (only 3 months after program launch), we found that the total number of problem records had increased to 45% (383/852), with 742 total problems. Of these, 113 records had late results.

In September 2017, a year after the introduction of the program, the

proportion of problem records decreased to 30% (282/955), including 479 total problems and 25 records with late results. Annual reports show continuing progress, so that by September 2020, the proportion of problem records decreased even further to 2% (32/1,304), with 63 total problems and 1 (0%) late result.

Next Steps

The program continues to focus on ensuring that records are maintained accurately and are up to date on ClinicalTrials.gov, including a zero-tolerance policy for late results. We review new registrations and record updates at regular intervals, conduct ongoing education in using the PRS, and update policies and standard operating procedures. We continue to evaluate tools (e.g., software) for managing the growing number of records.

As a CTSA hub, JHUSOM is driven to develop innovative solutions and increase efficiencies in clinical research. We have started monthly calls including other Johns Hopkins entities such as the Kennedy Krieger Institute and the School of Public Health to gain internal efficiencies. We plan to share the resources we have developed with our peer institutions through the taskforce, invited speaking engagements, and national meetings and publications, to enable us to contribute to national best practices.⁸

Future efforts will be to identify or develop software solutions to more efficiently distinguish trials funded by the NIH, detect records before they become problems, and track metrics over time.

To comply with the scientific, ethical, and legal requirements for registering, updating, and reporting the results of clinical trials on ClinicalTrials.gov, institutions need experts who stay current with guidelines, detect problems in the PRS, and provide training and support for investigators.⁹ Multiple offices, including IRBs¹⁰ and institutional leadership, must be engaged to secure resources to support these activities, champion the importance of research transparency, and administer consequences for noncompliance with existing regulations.

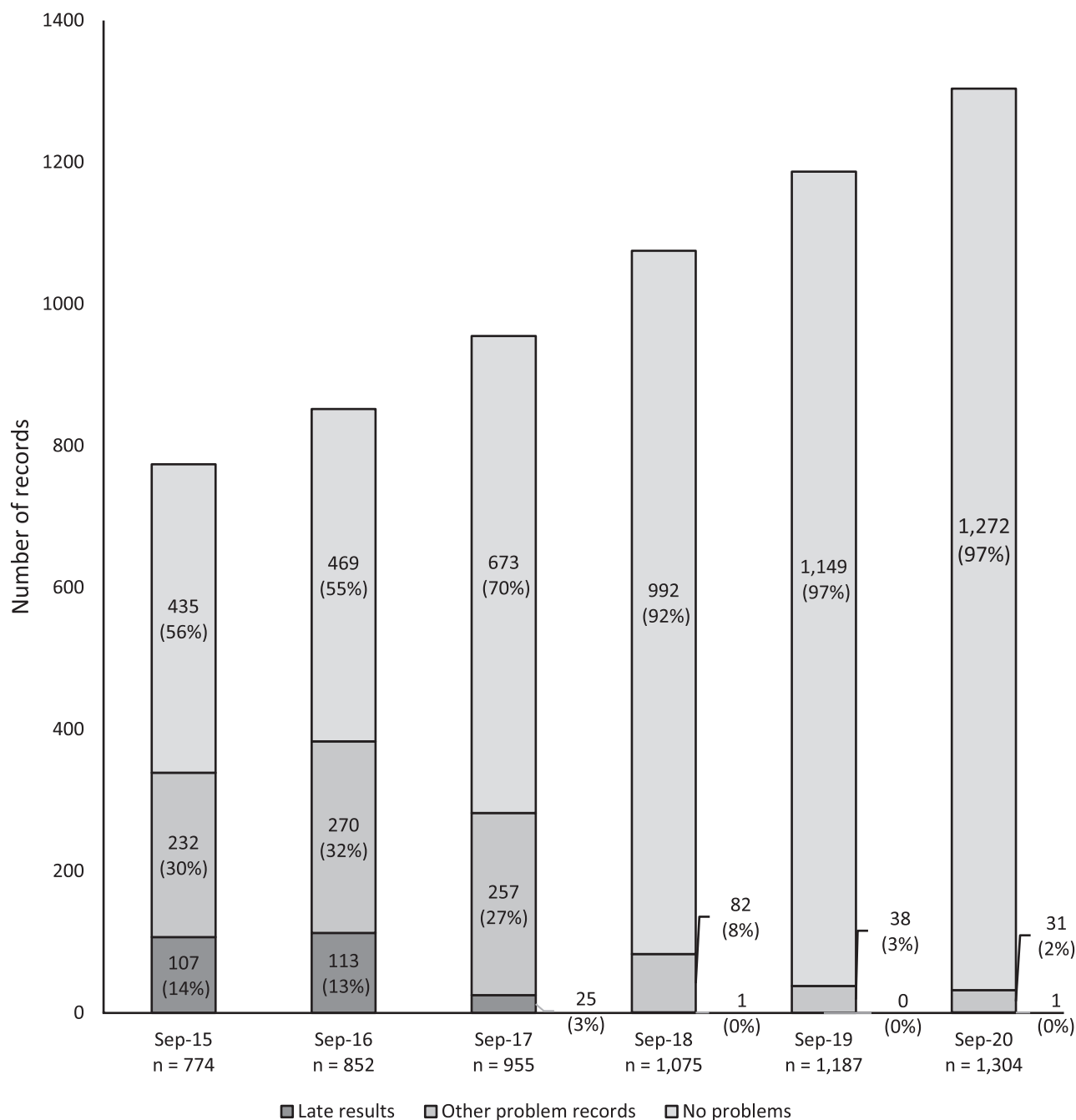


Figure 2 ClinicalTrials.gov records at Johns Hopkins University School of Medicine, 2015–2020. At program launch (2016), 45% of Johns Hopkins University records had problems, 13% with late results. Total number of records has grown each year; absolute number and proportion of records with problems have decreased steadily. In 2020, 2% of records had problems, 0% of records had late results.

Always strive for 100% compliance but realize there will inevitably be a few problems and occasional late results even for mature programs.

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Teaching and Learning Moments

The Art of Running and Medicine



“Why do you run?” the medical school interviewer asked as he scanned my application. It was a simple but profound question, and I struggled to articulate a response. His philosophical question resurfaced many times throughout medical school, and the answer gradually revealed itself on my journey to becoming a doctor. To my surprise, it was intimately linked with the human side of medicine that had so compelled me to enter the field.

During my first year, I noticed a trend that piqued my interest in running: A lot of trainees and physicians are runners. In one remarkable icebreaker during orientation week, I discovered that 6 out of the 10 medical students in my small group had completed a marathon. Selection bias was clearly at play, as individuals who choose a career in medicine often possess high degrees of both self-motivation and health consciousness. But discussions with my classmates yielded a more nuanced and satisfying theory. Running, we agreed, was like an invisible bridge between the mind and the body. Just as doctoring connects the pathophysiological and human elements of illness, running marries the physical and mental components of health.

One of the most memorable patients I met on the wards was a man who had

been paralyzed from the waist down after a motor vehicle accident. He had been an avid runner before the accident, and he said that learning to live without running was the most challenging part of his recovery. Sometimes he still had dreams where he was running a trail at sunrise and felt the sensation of absolute freedom. “These dreams are as real as running itself,” he told me, “and I cherish them.” Just as a cancer patient can paradoxically be frail of body but strong of spirit, this patient had seemingly broken the bonds of his paralytic illness, even if just for a fleeting moment. This encounter reinforced my conviction that running is a truly human experience. Similar to doctoring, it is impossible to tell the whole story with just textbook definitions.

When I scrubbed into my first surgery of medical school, I could not help but notice some uncanny similarities to running. I retracted for hours as my attending surgeon performed the total thyroidectomy. I was completely captivated by this new world of medicine where anatomy and physiology came to life before my eyes. Music played softly in the background and I had no concept of the passing time. I watched as the surgeon effortlessly tied off the blood vessels infiltrating the thyroid tumor, and I imagined that his mind was somewhere between intense focus and familiarity.

There is some evidence to suggest that the positive mood-altering effects of running are related to opioidergic pathways in the central nervous system. Although the mechanism is not yet fully understood, it lends support for a connection between the physical and mental aspects of running. As medical schools increasingly adopt wellness curricula and launch initiatives to destigmatize mental illness among trainees, running deserves a place in the conversation. It is important to recognize, however, that running is an intensely personal experience. What I find in running, others may find in yoga, weightlifting, swimming, boxing, guided meditation, or a plethora of other activities.

I recently ran into my medical school interviewer at a banquet hosted by the school alumni association. “I finally found the answer to your question about running,” I said, hoping he would remember our previous encounter. He looked at me, chuckled, and said, “I knew you would.” Then, he leaned in closer and whispered, “I was a runner too.”

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